E2 | Competitive Firms

Firms with no market power make zero profit.

Minutes	Animation or board work	Purpose
Warmup	Hold `0.warmup` before the timed lecture sequence	Retrieve average and marginal cost; identify the missing revenue information.
0–2	Spinach image and story (`0.opener`), predictions (`0.opener.question`), then bumper (`0.a`)	Establish the episode's question.
2–7	Play through `1.e.mr`	Market price, price taking, and why MR equals price.
7–9	Hold `1.f.exercise`; Q1a on board	Retrieve MR with a different price.
9–16	Resume through `2.e`	Profit, the next-unit comparison, and the chosen quantity.
16–21	Hold `2.f.exercise.b`, then `2.f.exercise.c`; work on board	Choose output and calculate profit.
21–28	Resume through `3.e`	Entry, losses, and operating versus shutting down.
28–32	Hide projection; Q2a	Compare two losses at a new price.
32–35	Resume through `3.f`	Long-run exit and zero economic profit.
35–37	Hide projection; Q2b–c	Distinguish exit from shutdown; retrieve entry.
37–47	Keep projection hidden; Q3	Apply the theory to the marginal-cost shock on the board.
47–49	Return to the opening question	Students explain the whole adjustment.
49–50	Resume to `4.a`	Preview downward-sloping firm demand and E3.

Exercise E1 | Warmup

Producing 40 units costs \$2,600. Producing 41 units costs \$2,681.

a) What is average total cost at 40 units?

b) How much does the 41st unit add to cost?

c) What else do we need to know to tell whether producing the 41st unit increases profit?

Opener

In 2006, an E. coli outbreak was linked to spinach.[1] California introduced a leafy-greens food-safety program a year later to make it harder for foodborne illness to spread through crops like spinach. Afterward, growers reported higher costs from this program. Suppose the extra cost is the same on every unit produced. What impact would a higher recurring cost have on the market?

Competitive Markets

Competitive markets have many buyers, many sellers, low barriers to entry, selling a homogeneous good, with no externalities.

We've just built up a model of the costs of production for the firm. From last time, the two most important cost curves for our firm are Average Total Cost (ATC) and Marginal Cost (MC).

Firms in competitive markets are just one of many. Since competitive firms like this are very small relative to the market, their choices have no real effect on the market. The firm graph essentially gives us a zoomed in view on the demand curve. And because we're so zoomed in, it doesn't have any meaningful slope.

This means competitive firms are what we call price takers. Changing quantity does not move the price in the market in any meaningful way.

So what does one more unit add to revenue? It sells for the market price, and we still get the same price for all the other units. That means the additional revenue from selling one more unit is the price. For a competitive firm, marginal revenue equals price.

Whiteboard | Exercise Q1a

At a price of 80, what does one more sale add to revenue? Why is MR the price?

[1]Source: UC Small Farms Network.

Profit and Output

At this price, what quantity maximizes the firm's profits? Profit is everything left over after we've paid our costs, which is TR - TC.

Profit is the difference between these two areas, so we usually just look at the difference on the graph.

This area is mathematically equivalent to Q(P-ATC), so we usually just show it like this.

If we start with a small quantity, we wonder whether we should increase quantity.

And as I hinted, profit will continue to rise until MR = MC, then it will fall.

Back to our question: How much should the firm choose to produce? At MC = MR!

As will likely feel familiar at this point in the semester, this is an example of marginal thinking. Our best decision involves thinking on the margin, thinking about the more or one less. When we choose based on whether we should change in little steps instead of big chunks, we can add clarity our decisionmaking.

Whiteboard | Exercise Q1.b–c

Choose output, then find profit.

Why don't we choose at the minimum of ATC?

Entry and Exit

At this price, the firm is making a profit, which is nice for them. But profit is nice for everyone. If you're looking at a market where others are making profit, what would *YOU* do? If we zoom back out to the whole market we can think about what happens to equilibrium when the firms are making profit.

The demand curve is just the willingness to pay of many people all added together. And the supply curve is the willingness to sell of many sellers all added together. We find equilibrium by intersecting the supply and demand curves. This is where the price for the firm comes from. But profit is nice

for everyone, so new firms open up to get in on some of that profit. New firms means the supply curve pivots out. More firms are willing to sell at each price.

You'll remember that a pivot out like this leads buyers and sellers to lower their price. And a lower price lowers profit for the firm. Firms will continue to enter until there is no profit left.

When price is low, firms are making a negative profit, or what we call a loss.

Firms earning a loss will shut down in the short run if price is below their AVC at the best positive output. Shutting down means producing nothing this period. We still have to pay the fixed cost. Exiting the industry is a different decision: it lets us avoid those costs in future periods.

The firm needs to decide whether operating with a loss is better than shutting down and still paying this period's fixed cost.

Here, the price is above average variable cost. By operating, the firm earns enough to cover its variable costs and part of its fixed cost. It loses 375 instead of the full fixed cost of 1000. So it keeps operating.

Whiteboard | Exercise Q2a

Compare operating with shutting down.

Long Run

In the long run, if price is below the firm's ATC, they're making a loss in the long run, and they exit, shifting the supply curve in, and the price up.

And if the price is above the firm's ATC, they're making a profit, and new firms enter, shifting the supply curve out, and the price down.

Combined, the only long-run stable position is when price is equal to ATC and MC, which also corresponds with zero profit. This gives us the zero profit condition: In the long run, firms earn zero profit.

Whiteboard | Exercise Q2b–c

Entry or exit, and the effect on price.

Whiteboard | Exercise Q3

~~Start at zero profit, raise fixed cost, and draw the immediate and long-run responses.~~ {++ Start at zero profit. Add the same cost to every unit, leaving fixed cost unchanged. Draw the short-run response, then the adjustment as firms exit. ++}

Resolve the episode question

~~Return to the opener: why do some firms leave while the survivors produce more?~~ {++ Return to the opener: what happens to the market when each unit costs more to produce? ++}

> ~~A recurring cost can make staying in the industry less attractive without making the next unit more expensive. Exit changes the market price, and that changes the remaining firms' production decisions. This is one mechanism that can matter when businesses face new compliance costs.~~ {++ If each unit costs more to produce, firms are willing to sell less at any given price. Market supply shifts in, price rises, and firms produce less. They make losses, so some leave. As firms exit, price rises further until the remaining firms cover their average total cost. With the same extra cost on every unit, survivors return to their original output, but fewer firms remain and the market produces less. ++}

Next time

Resume to `4.a`: "What happens to Marginal Revenue when demand slopes down?"